



Seaborn Visualization: Basics to Advanced

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Seaborn Basics

Introduction, scatter, line, histogram, box & bar plots

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Advanced Seaborn

Violin, pair, heatmap, joint & swarm plots + 3D graphs

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Plotly

Interactive scatter, box plot & scatter matrix

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Introduction to Seaborn

Seaborn is a **Python library for statistical data visualization** that builds on Matplotlib. It provides an easy interface for creating attractive and informative graphics.

Built-in Themes

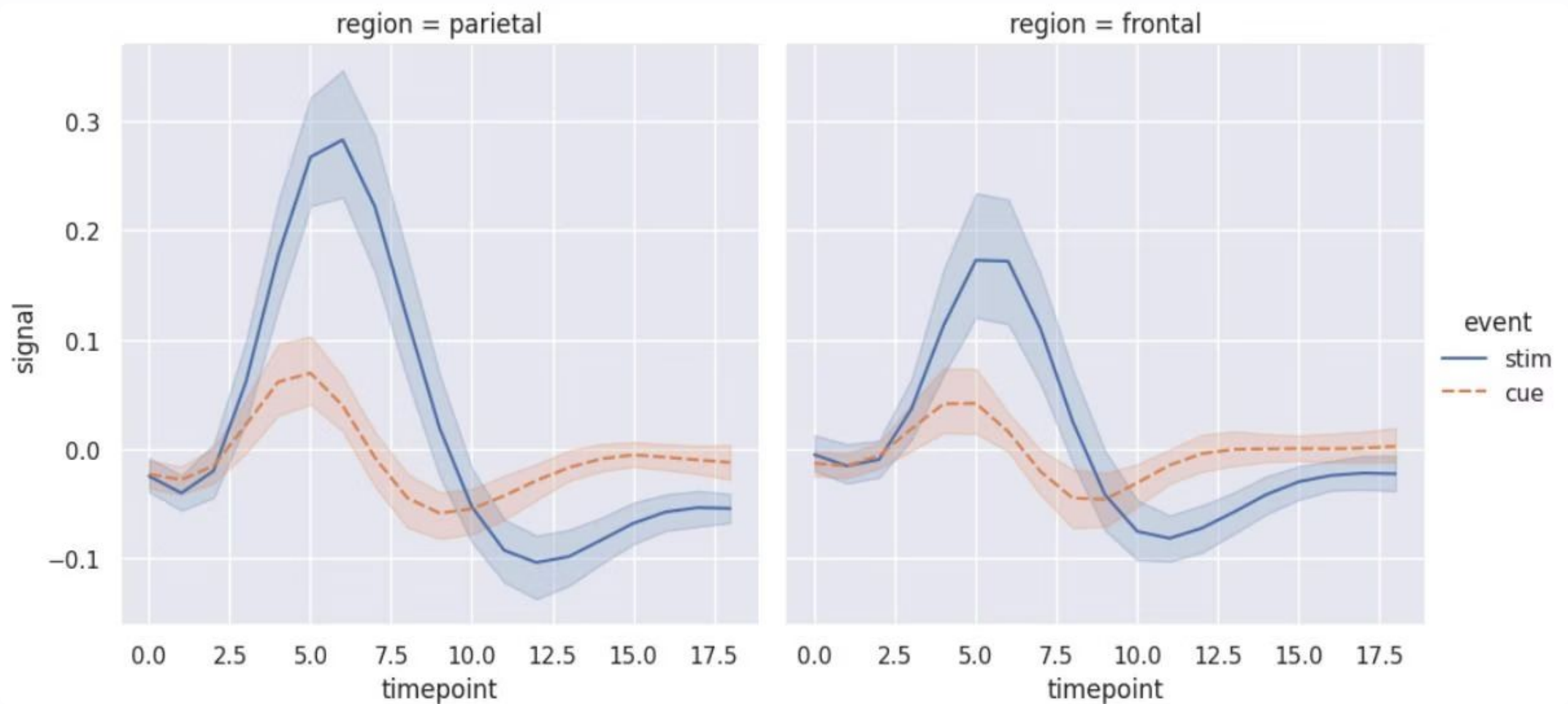
Beautiful color palettes and themes out of the box – no manual styling needed.

Complex Datasets

Especially suited for exploring datasets with multiple variables simultaneously.

Simpler Code

Achieves complex visualizations with fewer lines of code than Matplotlib.



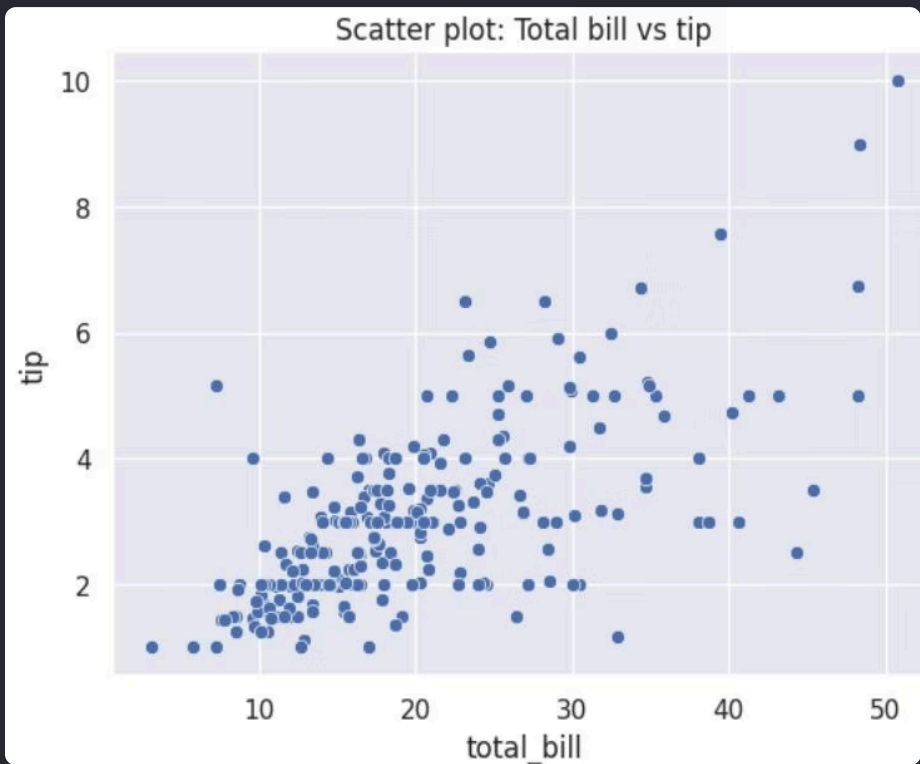
Example: Seaborn's

Plotting Graphs Using Seaborn

Seaborn achieves similar visualizations to Matplotlib but with **enhanced styling and simpler code**. Using the built-in `tips` dataset, here are four key plot types:

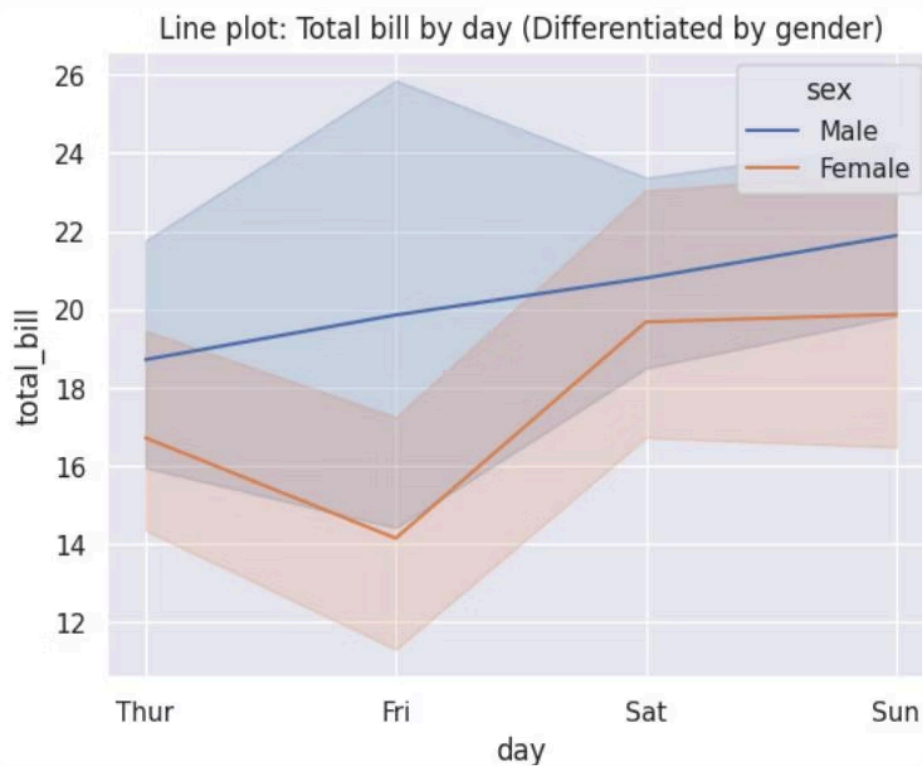
Scatter Plot

Shows a **positive relationship** between total bill and tip — customers tip more when they spend more. Other factors also influence tip variability.



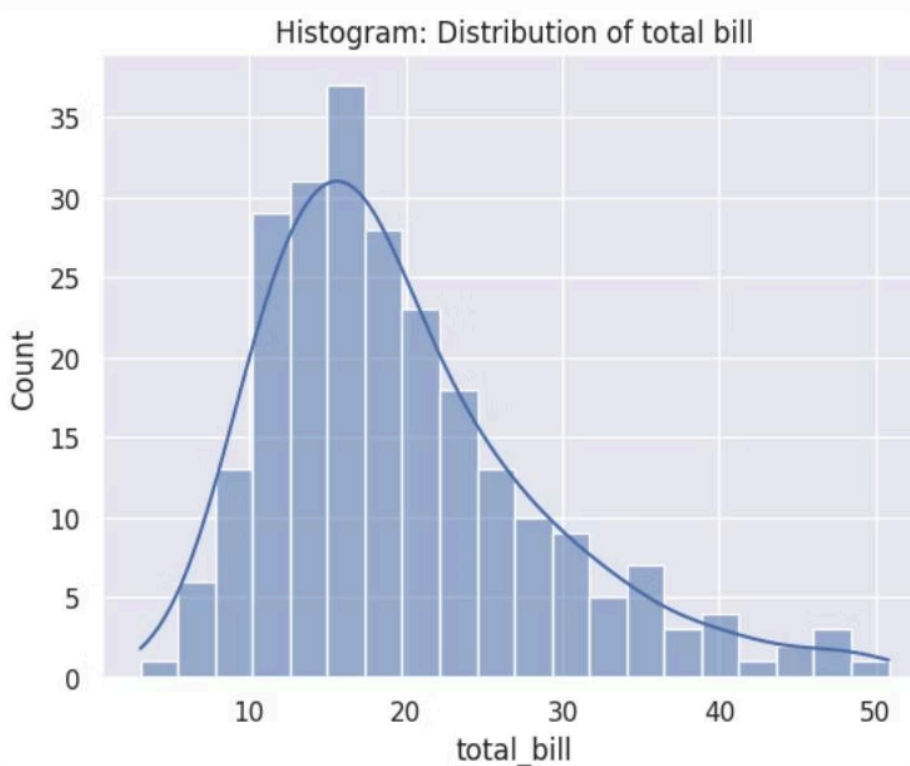
Line Plot (by Gender)

Males consistently pay **higher total bills** than females. The gap is most pronounced on **Friday** (Male ~26, Female ~14).



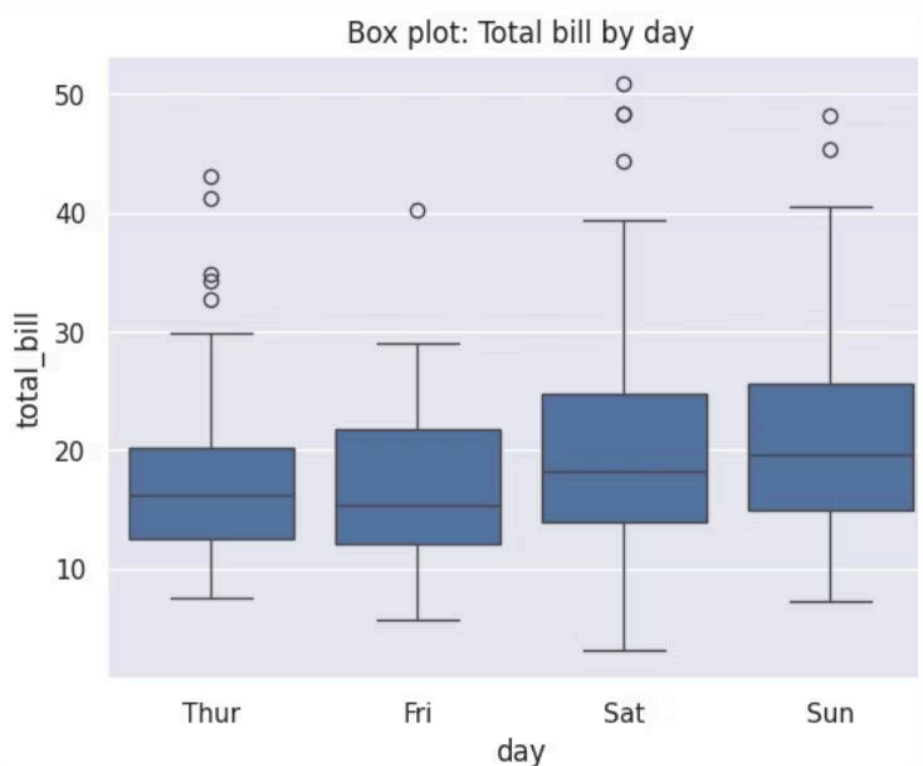
Histogram

Distribution of total bills is **right-skewed** — most bills fall in the **\$10–\$20 range**, with a KDE curve overlaid.

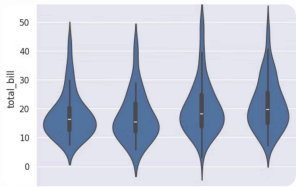


Box Plot & Bar Plot

Saturday & Sunday show higher median bills than Thu/Fri. Bar plot confirms males spend more, especially on **Thursday and Friday**.

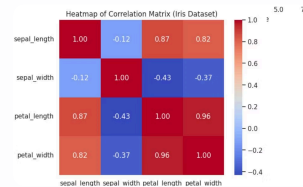


Advanced Seaborn: Violin, Pair, Heatmap, Joint & Swarm Plots



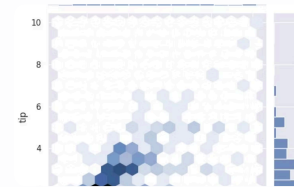
Violin Plot

Combines box plot + kernel density. Shows **data distribution shape** per day. Saturday has the highest median bill (~\$19).



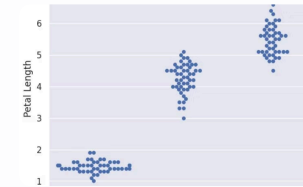
Heatmap

Visualizes **correlation matrix**. Iris dataset: petal_length vs petal_width = **0.96** (very strong). Warmer colors = stronger correlation.



Joint Plot

Hexbin scatter + marginal histograms. Highest density: bills \$10–\$20, tips \$1–\$3. Reveals bivariate distribution clearly.



Swarm Plot

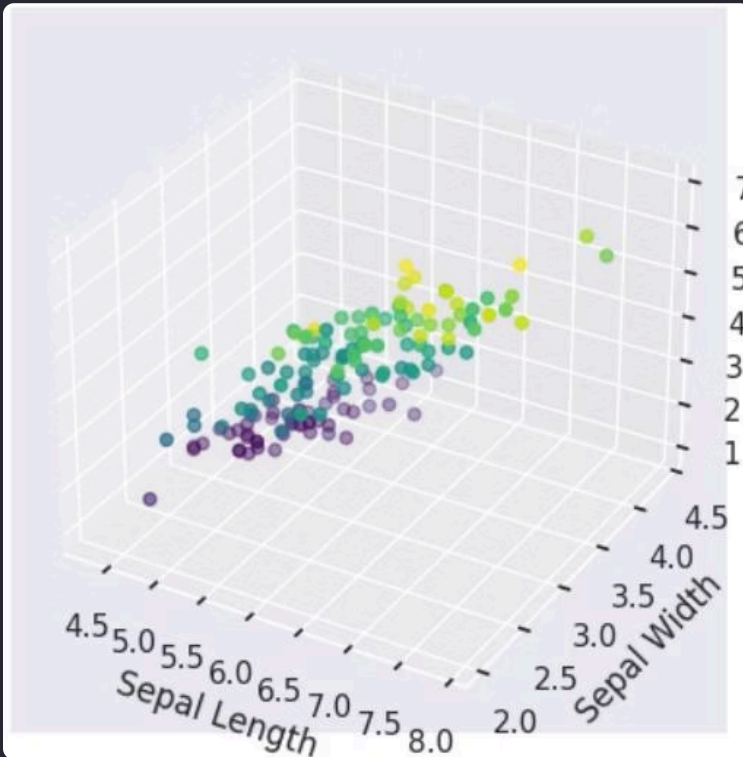
Categorical scatter — no overlapping points. Setosa: **1–2 cm**, Versicolor: **3–5 cm**, Virginica: **4.5–7 cm** petal length.

❏ Pair Plot (sns.pairplot): Shows all variable relationships at once. Males have higher total bills and tips than females across all combinations.

3D Graphs with Seaborn & Introduction to Plotly

3D Graphs Using Seaborn

Seaborn (with Matplotlib's 3D toolkit) can plot **3D scatter**, **helix**, and **surface plots** to visualize relationships among three variables simultaneously.



3D scatter of Iris: clear separation between species by petal length. Color encodes petal width (purple=low, yellow=high).

Introduction to Plotly

Plotly is a Python library for **interactive, web-based visualizations**. Key advantages over Seaborn:

- ☐ **Interactive**
Zoom, pan, hover over points to see values, toggle data series on/off.
- ☐ **Scatter & Box Plots**
Plotly scatter: Iris species form distinct clusters. Box plot: Virginica has highest petal length (median ~5.6 cm).
- ☐ **Scatter Matrix**
Like Seaborn's pairplot — shows all pairwise variable relationships, colored by species. Quickly identifies correlations and patterns.

Seaborn

Statistical, beautiful, simple code — best for exploration

Plotly

Interactive, web-ready — best for presentations & dashboards

Together

Use both for a complete data visualization workflow